

FAULT CODE 286 - SAE J1939 Multiplexing Configuration Error - Out of Calibration

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an active fault code.	Fault Code 286 active?
STEP 2.	Check the Original Equipment Manufacturer (OEM) vehicle electronic control unit and Engine Control Module (ECM) for proper multiplexing configuration.	
	STEP 2A. Determine which vehicle electronic control unit multiplexed components (switches, accelerators, or sensors) are enabled for multiplexing over the J1939 data link to the ECM and compare to the ECM configuration.	Does the ECM multiplexing configuration match the OEM vehicle electronic control unit multiplexing configuration?
STEP 3.	Check ECM calibration and clear fault codes.	
	STEP 3A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 3B. Disable the fault code.	Fault code inactive?

STEP 1. Check the fault codes.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for an active fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 286 active? YES	2A
	Fault Code 286 active? NO Repair : An OEM issue has been detected. Contact the OEM for further repair instructions. It is still possible that there is a vehicle electronic control unit, vehicle electronic control unit setup, or vehicle electronic control unit to data link connection issue.	Repair complete.

STEP 2. Check the Original Equipment Manufacturer (OEM) vehicle electronic control unit and Engine Control Module (ECM) for proper multiplexing configuration.

STEP 2A. Determine which vehicle electronic control unit multiplexed components (switches, accelerators, or sensors) are enabled for multiplexing over the J1939 data link to the ECM and compare to the ECM configuration.

Conditions :

- Determine proper OEM vehicle electronic control unit multiplexing configuration component enables and OEM vehicle electronic control unit source addresses from the appropriate OEM information, or from a saved job image.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Check the multiplexing configuration. Use INSITE™ electronic service tool SAE J1939 Multiplexed Fault Data, located in Advanced ECM data, to determine which multiplexed component is causing the fault. If the status column indicates "Active" for a multiplexed component, check that the ECM multiplexed component enables and source addresses match the OEM vehicle electronic control unit multiplexed component enables and source addresses. These components can be found under SAE J1939 Multiplexing in Features and Parameters. Use the following for OEM specific multiplexing configuration information. Refer to Service Bulletin, Multiplexing Troubleshooting, Bulletin 4021378 (/qs3/pubsys2/xml/en/bulletin/4021378.html).	Does the ECM multiplexing configuration match the OEM vehicle electronic control unit multiplexing configuration? YES Repair : An OEM issue has been detected. Contact the OEM for further repair instructions. It is still possible that there is a vehicle electronic control unit, vehicle electronic control unit setup, or vehicle electronic control unit to data link connection issue.	3A
	Does the ECM multiplexing configuration match the OEM vehicle electronic control unit multiplexing configuration? NO Repair : An incorrect setup has been detected in the Cummins® ECM. Enable the proper components for multiplexing on the applicable OEM application and make sure the OEM vehicle electronic control unit source addresses for each component are correct.	3A

STEP 3. Check ECM calibration and clear fault codes.

STEP 3A. Check if an ECM calibration update is available.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. • Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Data Plate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	3B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	3B

STEP 3B. Disable the fault code.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code inactive? YES	Repair complete.
	Fault code inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A